Supplementary Material for "Trust Dynamics in Multi-Agent Strategic Interaction: A Simulation Study of Bayesian Reputation Systems in 8-Player Limit Texas Hold'em"

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S1. Mathematical preliminaries

Two quantitative tools recur in every part of the analysis, and we state them here so the main text can use them without further build-up.

Pearson correlation coefficient is the first one. Given n paired observations (x_i, y_i) with sample means $\bar{x}$ and $\bar{y}$, it is defined as

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}. \tag{1}$$

r ranges from -1 (perfectly anti-correlated) through 0 (no linear relationship) to $+1$ (perfectly correlated); the sign indicates the direction of the relationship and the magnitude its strength. We use r between an archetype's mean trust score and its final stack to measure how reputation maps to economic outcome—a value we call r_{tp}.

Bayesian updating is the second. Given a prior belief $P(t)$ that the opponent is of type t, and a likelihood $P(a \mid t)$ that a type-t opponent would have taken the action a we just observed, Bayes' rule gives the posterior belief

$$P(t \mid a) = \frac{P(a \mid t)\,P(t)}{\sum_{t'} P(a \mid t')\,P(t')}. \tag{2}$$

Each agent maintains a categorical posterior distribution over the 8 archetypes for each other player at the table. They update their beliefs with Eq. 2 on every action they observe. The belief-updating mechanism used in this study—including the noise floor and hand-forgetting—is developed in the main text, as is the honesty score $h(t)$ that converts a posterior into a scalar trust score.

S2. Additional tables, figures, and hand transcripts

Supplementary Box S1. An instance of the trap (Phase 3 hand #67, 30 actions, 110-chip pot)

Setup. Seed 42, hand 67. **Wall** (seat 3) is dealt 2♠ 5♣ – rank 6,749, the worst hand at the table. Final pot 110 chips, showdown.

Action sequence:

Seq	Seat	Round	Action	Amt	Pot	Notes
3	**Judge**	preflop	raise	4	7	
6	**Firestorm**	preflop	raise	6	13	3-bet
7	**Wall**	preflop	call	5	18	*calls 3-bet pot with 2-5 off*
9	**Judge**	preflop	call	2	20	
10–16	…	flop	…		38	**Judge** bets, **Firestorm** raises, **Wall** calls each street
17–23	…	turn	…		74	Same pattern
24–30	…	river	…		110	**Wall** calls the final two raises

Showdown. Firestorm J♡ J♢ (rank 4042) **wins** 110. **Judge** 8♡ 8♢ (rank 4703). **Wall** 2♠ 5♣ (rank 6749). *Wall pays 35 chips into a hand it had no chance to win. Every observer's posterior saturates near its ceiling—just short of* 1*, given the* ε *noise floor—on* ***Firestorm*** *being a firestorm and on* ***Wall*** *being a wall.*

TABLE S1

A PRIMER ON FIXED-LIMIT TEXAS HOLD'EM. THE MINIMUM SET OF CONCEPTS NEEDED TO READ THE REST OF THE PAPER. THE BETTING BOUNDS IN THE "BET SIZING" ROW ARE THE SPECIFIC VALUES WE USE THROUGHOUT THE SIMULATION; EVERYTHING ELSE APPLIES TO FIXED-LIMIT HOLD'EM IN GENERAL.

Concept	Description
Hole cards	The two private cards each player is dealt at the start of every hand; only that player can see them.
Community cards	Up to five face-up cards shared by everyone at the table; players combine these with their hole cards to form a 5-card hand.
Betting rounds	Four sequential rounds per hand: preflop (no community cards yet), flop (three community cards revealed), turn (one more), and river (the fifth and final).
Actions	At each turn a player may fold (give up the hand), check (pass without betting, only if no one has bet), call (match the current bet), bet (open the wagering), or raise (increase an existing bet).
Bet sizing (this paper)	Small bet 2 chips on preflop and flop; big bet 4 chips on turn and river; no more than four raises per round (the "four-bet cap").
Blinds	Forced bets: small blind 1 chip and big blind 2 chips, posted by the two seats clockwise of the dealer and rotating each hand.
Pot	The accumulated chips at stake in a hand. The winner of the hand collects the pot.
Showdown	End-of-hand reveal: surviving players turn over their hole cards and the strongest 5-card combination wins. A hand that ends with everyone folding never reaches showdown.
Stack	The chips a player has in front of them; depleting to zero triggers a rebuy in our setup.

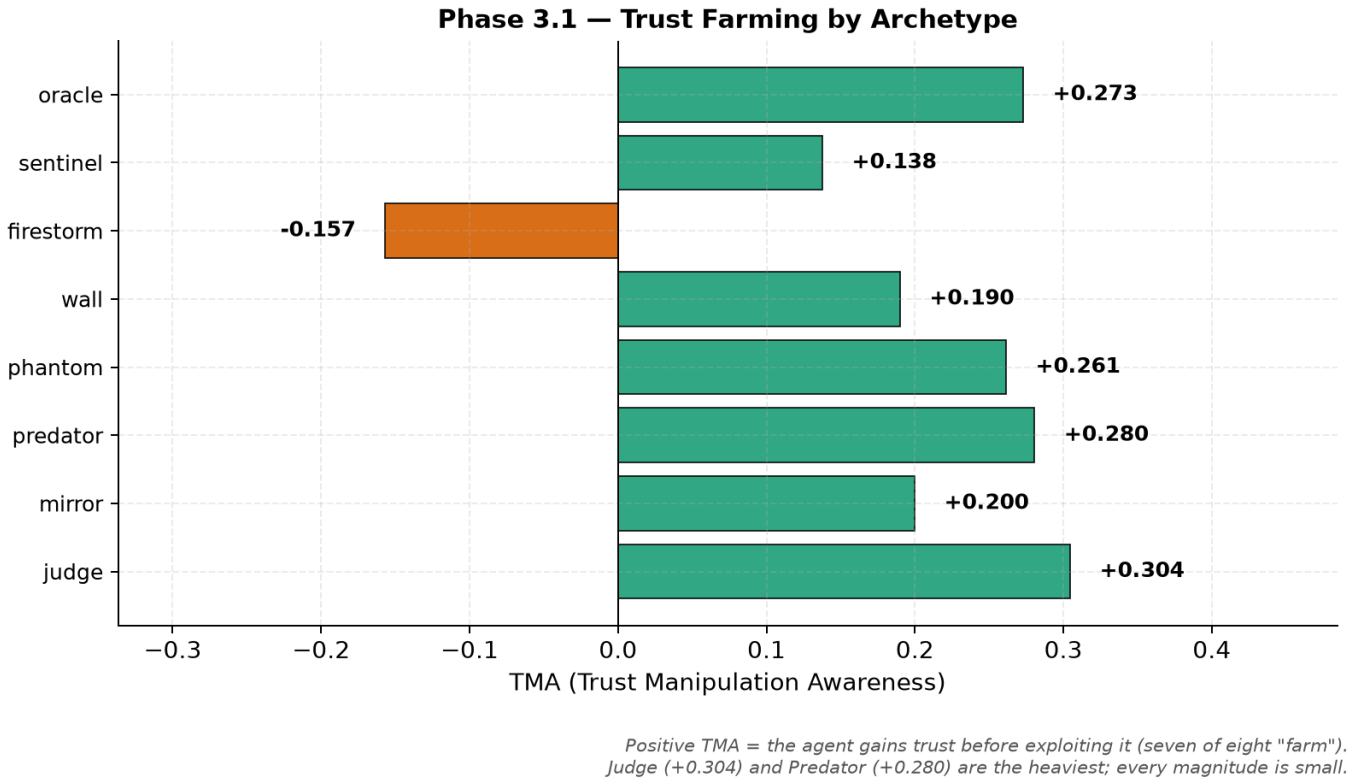


Fig. S1. **Trust Manipulation Awareness (TMA) per archetype in Phase 3.1.** Positive TMA indicates an agent's rolling trust delta correlates with its rolling aggression delta—building trust through cooperative early play, then leveraging it for higher-aggression later moves. Seven of eight archetypes show weakly positive TMA; **Judge** (+0.30) and **Predator** (+0.28) are the strongest.

Supplementary Box S2. The inversion in a single hand (original five-seed Phase 3.1 exploration, hand #146, 17 actions, 32-chip pot)

Setup. Seed 42, hand 146. **Wall** (seat 3) is dealt $K\heartsuit\ Q\heartsuit$ – rank 872, a strong hand. Final pot 32 chips, showdown.

Action sequence:

Seq	Seat	Round	Action	Amt	Pot	Notes
7	**Firestorm**	preflop	raise	3	10	
8	**Wall**	preflop	call	2	12	calls 3-bet with K-Q suited
11	**Firestorm**	flop	bet	2	14	c-bet
12	**Wall**	flop	call	2	16	
13	**Firestorm**	turn	bet	4	20	second barrel
14	**Wall**	turn	call	4	24	
15	**Firestorm**	river	**check**	0	24	*tell — bluff line abandoned*
16	**Wall**	river	**bet**	4	28	*value bet against checked range*
17	**Firestorm**	river	call	4	32	

TABLE S2

PER-ARCHETYPE POLICY BOUNDS. $P(\text{BET}|S)$, $P(\text{BET}|W)$ ARE THE PROBABILITIES OF BETTING WITH A STRONG OR WEAK HAND ON A GIVEN STREET (PREFLOP UNLESS NOTED); $P(\text{FOLD}|W)$ IS THE PROBABILITY OF FOLDING A WEAK HAND TO AN OPPONENT'S BET. STATIC ARCHETYPES LIVE AT A FIXED POINT INSIDE THE BOUND BOX. UNDER PHASE 2'S BOUNDED HILL-CLIMBER THE AGENT MAY DRIFT WITHIN THE BOX; UNDER PHASE 2*'S UNBOUNDED VARIANT THE BOX IS REPLACED BY $[0, 1]$ ON EVERY DIMENSION. ADAPTIVE ARCHETYPES ADDITIONALLY READ LIVE STATE (OPPONENT POSTERIOR OR GRIEVANCE LEDGER) AND ARE DESCRIBED BY THE RULE THAT PRODUCES THE POLICY RATHER THAN BY FIXED BOUNDS. THE BOUND BOXES DESCRIBE EACH ARCHETYPE'S *policy*; THE OBSERVER'S LIKELIHOOD MODEL INSTEAD READS FOUR SUMMARY STATISTICS PER ARCHETYPE (BLUFF RATE, VALUE-BET RATE, CALL RATE, MADE-BET RATE). FOR **SENTINEL** AND **JUDGE**'S COOPERATIVE STATE THOSE FOUR ARE BYTE-IDENTICAL AT $(0.083, 0.900, 0.325, 0.225)$ DESPITE THE DIFFERING BOXES, WHICH IS THE IDENTIFIABILITY FLOOR DISCUSSED IN THE MAIN TEXT.

Archetype	$P(\text{bet}\|S)$	$P(\text{bet}\|W)$	$P(\text{fold}\|W)$	h	Adaptation rule
Oracle	$[0.65, 0.85]$	$[0.15, 0.35]$	$[0.45, 0.65]$	0.67	None — fixed mixed strategy
Sentinel	$[0.90, 1.00]$	$[0.00, 0.10]$	$[0.80, 0.95]$	0.92	None — tight value-bettor
Firestorm	$[0.85, 1.00]$	$[0.45, 0.65]$	$[0.00, 0.20]$	0.38	None — bets everything
Wall	$[0.95, 1.00]$	$[0.00, 0.05]$	$[0.00, 0.05]$	0.96	None — calls all, folds none
Phantom	$[0.30, 0.50]$	$[0.60, 0.80]$ pre, $[0.05, 0.20]$ post	$[0.70, 0.90]$ post-flop after preflop bluff	0.48	Static but street-conditional
Predator	TAG default; exploit profile vs. target	0.05 default; 0.55 vs. classified target	0.65 default; 0.30 vs. classified target	0.79	Activates exploit profile when posterior on opponent > 0.60
Mirror	$0.5 \cdot \text{self} + 0.5 \cdot \text{target}$	same blend	same blend	0.91	Re-targets at each hand to highest-VPIP opponent
Judge	$[0.88, 0.98]$ coop; $[0.95, 1.00]$ retal.	$[0.02, 0.10]$ coop; $[0.30, 0.45]$ retal.	$[0.78, 0.92]$ coop; $[0.30, 0.50]$ retal.	0.92	Switches to retaliation after 5+ confirmed bluffs by an opponent

Archetype	TMA	Direction
judge	+0.304	farming
predator	+0.280	farming
oracle	+0.273	farming
phantom	+0.261	farming
mirror	+0.200	farming
wall	+0.190	farming
sentinel	+0.138	farming
firestorm	-0.157	no awareness

TABLE S3

TRUST MANIPULATION AWARENESS (TMA) PER ARCHETYPE IN PHASE 3.1 (TWENTY-SEED TEMPERATURE-0 REPLICATION). POSITIVE TMA = THE AGENT GAINS TRUST BEFORE EXPLOITING IT. SEVEN OF EIGHT ARCHETYPES SHOW WEAKLY POSITIVE TMA (MAXIMUM +0.30); FOR HONEST ARCHETYPES THE SIGNAL REFLECTS CORRELATED TRUST AND LEGITIMATE VALUE-BETTING RATHER THAN DECEPTION.

Archetype	VPIP P1	VPIP P3.1	PFR P1	PFR P3.1	AF P1	AF P3.1
oracle	0.216	0.304	0.061	0.099	1.18	1.16
sentinel	0.149	0.116	0.083	0.091	2.12	5.04
firestorm	0.494	0.741	0.120	0.473	1.12	3.33
wall	0.539	0.902	0.015	0.039	0.13	0.22
phantom	0.279	0.326	0.092	0.091	1.45	1.31
predator	0.215	0.193	0.063	0.103	1.20	1.99
mirror	0.214	0.177	0.060	0.091	1.18	1.61
judge	0.149	0.158	0.083	0.091	2.10	2.14

TABLE S4

PHASE 1 (FROZEN RULES) VS. PHASE 3.1 (LLM + REASONING) BEHAVIORAL FINGERPRINTS. PHASE 1 COLUMNS ARE THE FIVE CANONICAL SEEDS AT 10 000 HANDS; PHASE 3.1 COLUMNS ARE THE TWENTY-SEED TEMPERATURE-0 REPLICATION AT 150 HANDS PER SEED. VPIP = VOLUNTARILY PUT IN POT; PFR = PREFLOP RAISE RATE; AF = AGGRESSION FACTOR ((BETS+RAISES)/CALLS).

BROADER IMPACT STATEMENT

This research examines ways in which an observation-based reputation system creates incentives for an agent to exploit its predictability with respect to cooperative behavior. Therefore, the mechanisms identified in this work can be used to develop a theoretical basis for discovering, modeling, or strategically leveraging trust in reputation systems. In particular, both the Trust Manipulation Awareness (TMA) metric and the concept of "trust farming" can be viewed as a conceptual approach for maximizing the use of trust instead of detecting it.

However, we believe that the practical risk of such misuse is limited by the scope of the findings presented here. All of our experiments were conducted in a synthetic, zero-sum game, and we continually argue throughout the paper that we should not infer direct application to real-world, positive-sum reputation systems. Instead, our results describe qualitatively, at a mechanism level, what appears to happen under the conditions studied rather than providing a specific deployable tactic against any existing marketplace, credit, or social reputation system. Each of those systems contains built-in institutional and enforcement mechanisms which were omitted from our testbed.

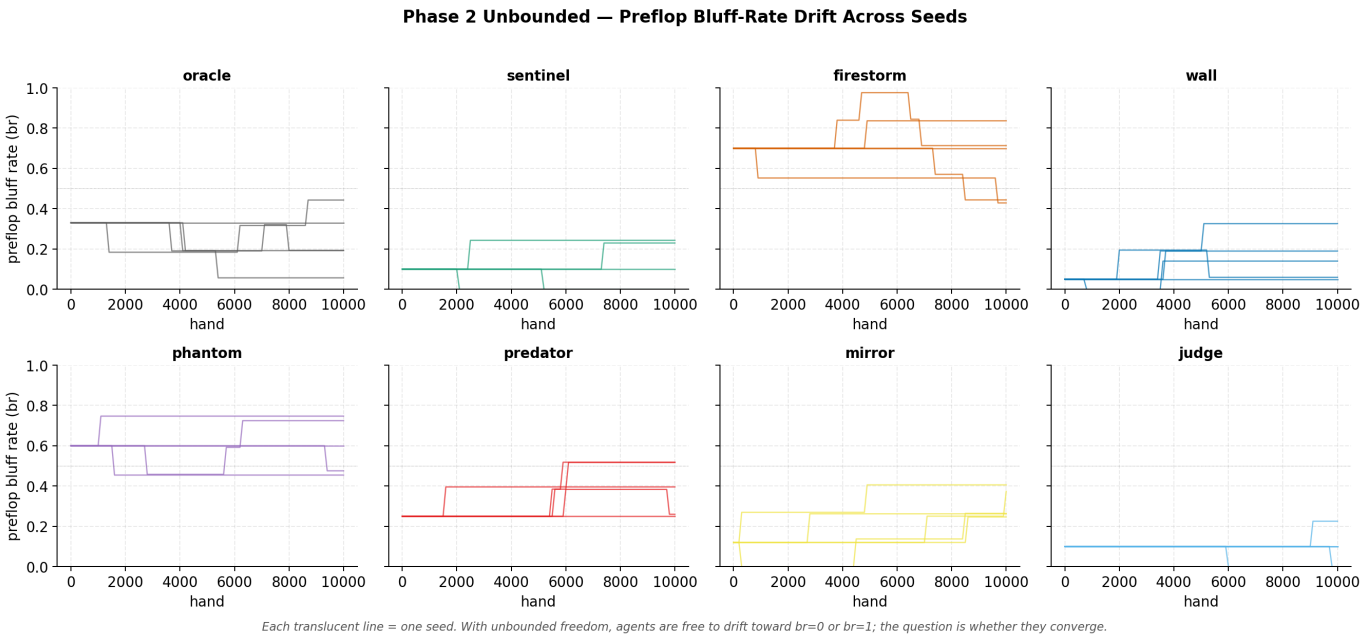


Fig. S2. **Per-archetype preflop bluff-rate drift under aggressive unbounded hill-climbing.** Each panel is one archetype and each translucent line is one of the five seeds. Every archetype's bluff rate wanders substantially over the run, yet the panels do not converge on a common profile (the corresponding mean L_1 parameter drift is 3.4).

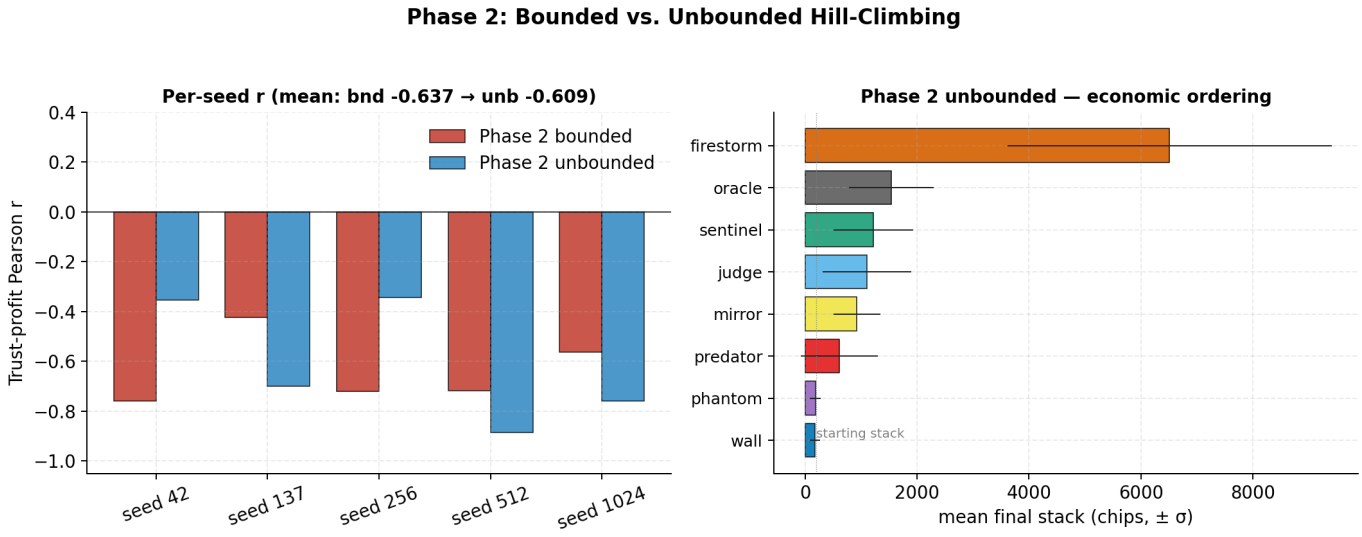


Fig. S3. **Bounded versus unbounded hill-climbing per seed.** The left of each pair of bars is r_{tp} for bounded Phase 2 hill-climbing; the right is r_{tp} for aggressive Phase 2* hill-climbing, for each of the five canonical seeds. When the bounding constraints are relaxed and the optimiser's step size is increased fivefold, r_{tp} increases by only $+0.028$ on average, and across seeds the variance in r_{tp} nearly triples.

It is necessary to understand how an agent's ability to act cooperatively can be exploited in order to build systems that will resist such exploitation. Our results indicate that observation-based reputation systems have blind spots where agents possess overlapping behavioral profiles, and that defending cooperative agents may necessitate developing defense systems capable of reasoning beyond merely increasing the amount of observational information available. We thus consider this work as contributing toward understanding and mitigating mechanisms for exploiting trust as opposed to being an instrument for performing such exploitation. To facilitate greater scrutiny and replication, all experiments are fully specified and seeded, and the source code and raw data are available for public review.

Supplementary Materials

The following supporting materials are available in the project repository:

1) **Bayesian update worked example.** `docs/worked_examples.md`

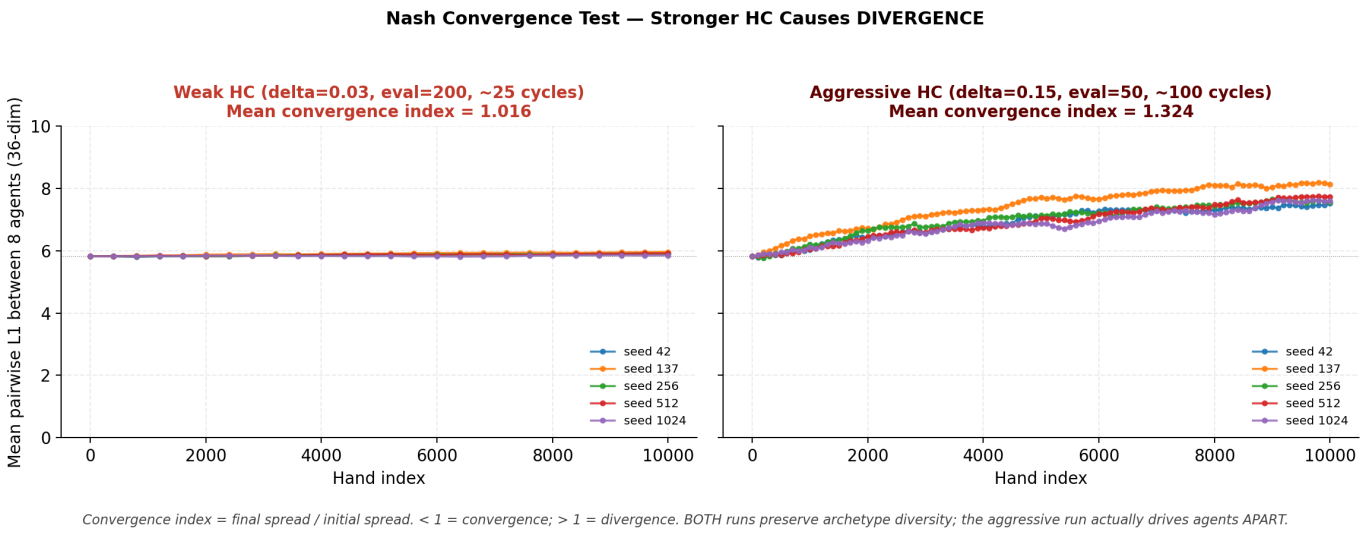


Fig. S4. **The 8-agent population becomes dispersed in parameter space.** Averaged pairwise L_1 distance between all eight agents in a 36-dimensional parameter space over 10,000 hands of unbounded hill-climbing. (**Left**) With weak HC ($\delta = 0.03$), agents barely move. (**Right**) Aggressive HC ($\delta = 0.15$) spreads the population out on every seed. A ratio of final-to-initial cluster spread > 1 implies the agents become more dispersed at end-of-run than at the start.

2) **Identifiability ceiling derivation.** `docs/stage5_identifiability.md`
3) **Phase 2* unbounded-optimization methodology.** `paper_resources/notes/phase2_unbounded_writeup_aggressive.md`
4) **Methods disclosures** (sampling temperature, side-pot construction, confidence intervals). `paper_resources/notes/methods_disclosures.md`
5) **Bootstrap and Fisher-z CI table.** `paper_resources/data/r_bootstrap_ci.csv`